

Note on: *On the pricing of Credit-Linked Notes: Evidence
from the Swedish market*

Marc Axelsson & Tobias Renström (SSE), 2013

Reading notes

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Authors	Marc Axelsson, Tobias Renström
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One-paragraph summary

A master's thesis that builds a reduced-form pricing model for credit-linked notes (CLNs) and applies it to 11 Nordea-issued CLNs in the Swedish market (2012–2013). The valuation is a survival-weighted coupon bond (1) with floating coupons set via forward rates and survival probabilities bootstrapped from CDS spreads via the piecewise-flat hazard rate model of O’Kane & Turnbull (2003). Counterparty credit risk (issuer Nordea) is layered on top via a bivariate Gaussian copula on the joint survival of the reference entity and the issuer, with default correlations estimated from 5Y CDS-spread time series. Empirically: all 11 CLNs are overpriced relative to the model, by 0.6%–6.4% under the baseline 40% recovery and up to 8.5% once issuer (Nordea) counterparty risk is priced in. The theoretical price tracks the 5Y CDS spread on the reference name almost perfectly, validating the model’s accuracy. The authors attribute the residual mispricing to retail-investor willingness-to-pay, possible “too-big-to-fail” perception of the issuer, and limited liquidity. Yield-curve interpolation uses Nelson-Siegel-Svensson on STIBOR + Swedish swap zero curve.

1 Setup and notation

Paper	House	Meaning
V^{CLN}	—	Theoretical CLN value at valuation date t
C	—	Coupon rate (in body: fixed; in practice STIBOR + spread)
F	—	Notional / face value
R	Rec	Recovery rate (baseline 40%)
T_i	T_i	Coupon payment dates ($i = 1, \dots, N$)
N	N	Number of coupons (4 quarterly, then yearly to maturity)
$Z(t, T_i)$	D_{t, T_i}	Risk-free / STIBOR discount factor
$Q(t, T_i)$	$Q(t, T_i)$	Survival probability of the reference entity
f_{t_1, t_2}	f_{t_1, t_2}	Forward rate, eq. 2
λ	λ	Hazard rate (piecewise-flat)
$S(t_0, t_N)$	—	CDS contractual spread
$\Delta(t_{n-1}, t_n)$	α_n	Day-count fraction in the appropriate convention
M	—	Discretisation steps per year for the protection leg
Q_{rp}^C	—	Joint reference-issuer survival via Gaussian copula
ρ	ρ	Default correlation (estimated from 5Y CDS correlations)
Φ, Φ_Σ	—	Univariate / bivariate standard normal CDF

2 Key equations

CLN value (risk-free issuer):

$$V^{CLN} = \left[\sum_{i=1}^N Z(t, T_i) Q(t, T_i) C + Z(t, T_N) Q(t, T_N) \right] F + \sum_{i=1}^N Z(t, T_i) (Q(t, T_{i-1}) - Q(t, T_i)) R F. \quad (1)$$

Says: present value of coupons (risk-discounted by both Z and Q) plus the survival-weighted notional plus a recovery payment on default in each interval. Structurally identical to a defaultable coupon bond with constant recovery (Bomfim 2005).

Forward rate for floating coupons:

$$f_{t_1, t_2} = \left(\frac{(1 + r_2)^{t_2 - t_0}}{(1 + r_1)^{t_1 - t_0}} \right)^{1/(t_2 - t_1)} - 1. \quad (2)$$

Says: Fabozzi–Mann forward rate from observed spot rates; r_1, r_2 are spot rates to t_1, t_2 . The fixed CLN spread b is added on top of f to get the per-period coupon.

Reduced-form hazard model (Jarrow–Turnbull):

$$\mathbb{P}[\tau < t + dt \mid \tau \geq t] = \lambda(t) dt, \quad Q(t_V, T) = \exp\left(-\int_{t_V}^T \lambda(s) ds\right). \quad (3)$$

Says: default times follow an inhomogeneous Poisson process with deterministic intensity $\lambda(t)$, assumed independent of interest rates and recovery.

CDS premium leg (with accrued):

$$\text{Premium PV} = S(t_0, t_N) \sum_{n=1}^N \Delta(t_{n-1}, t_n) Z(t_V, t_n) \left[Q(t_V, t_n) + \frac{1}{2} (Q(t_V, t_{n-1}) - Q(t_V, t_n)) \right]. \quad (4)$$

Says: O’Kane-Turnbull approximation: the half-period accrued premium on default is added to the full-period premium on survival to coupon date.

CDS protection leg:

$$\text{Protection PV} = (1 - R) \sum_{m=1}^{M \times t_N} Z(t_V, t_m) (Q(t_V, t_{m-1}) - Q(t_V, t_m)). \quad (5)$$

Says: default-leg integral discretised at M steps per year. Discretisation error of order $r/(2M)$; at $M = 4$, $r = 2\%$, 500 bp CDS spread, error $\approx 0.25\%$ relative or 1.25 bp absolute.

CDS breakeven (par) spread:

$$S(t_V, t_N) = \frac{(1 - R) \sum_{m=1}^{M \times t_N} Z(t_V, t_m) [Q(t_V, t_{m-1}) - Q(t_V, t_m)]}{\sum_{n=1}^N \Delta(t_{n-1}, t_n) Z(t_V, t_n) [Q(t_V, t_n) + \frac{1}{2} (Q(t_V, t_{n-1}) - Q(t_V, t_n))]} \quad (6)$$

Says: sets premium PV = protection PV. The denominator is the risky annuity (with accrued-on-default correction).

Piecewise-flat hazard term structure (bootstrap target):

$$Q(t_V, T) = \begin{cases} e^{-\lambda_{0,1} \tau} & 0 < \tau \leq 1 \\ e^{-\lambda_{0,1} - \lambda_{1,3}(\tau-1)} & 1 < \tau \leq 3 \\ e^{-\lambda_{0,1} - 2\lambda_{1,3} - \lambda_{3,5}(\tau-3)} & 3 < \tau \leq 5 \\ e^{-\lambda_{0,1} - 2\lambda_{1,3} - 2\lambda_{3,5} - \lambda_{5,7}(\tau-5)} & 5 < \tau \leq 7 \\ e^{-\lambda_{0,1} - 2\lambda_{1,3} - 2\lambda_{3,5} - 2\lambda_{5,7} - \lambda_{7,10}(\tau-7)} & \tau > 7 \end{cases} \quad (7)$$

where $\tau = T - t_V$ and $\lambda_{i,j}$ is bootstrapped from CDS quotes at maturities $\{1, 3, 5, 7, 10\}Y$.

Nelson-Siegel-Svensson yield curve:

$$y(\tau) = \beta_1 + \beta_2 \left(\frac{1 - e^{-\lambda_1 \tau}}{\lambda_1 \tau} \right) + \beta_3 \left(\frac{1 - e^{-\lambda_1 \tau}}{\lambda_1 \tau} - e^{-\lambda_1 \tau} \right) + \beta_4 \left(\frac{1 - e^{-\lambda_2 \tau}}{\lambda_2 \tau} - e^{-\lambda_1 \tau} \right). \quad (8)$$

Says: parametric zero-curve fit on STIBOR + Swedish swap zero quotes; the same form the Swedish Riksbank uses.

Gaussian copula joint survival (counterparty risk):

$$Q_{rp}^C(\tau, T) = \Phi_{\Sigma} \left(\Phi^{-1}(e^{-\lambda_p(T-\tau)}), \Phi^{-1}(e^{-\lambda_r(T-\tau)}); \rho \right), \quad \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}. \quad (9)$$

Says: the joint survival of reference r and issuer counterparty p is obtained by inverting the marginal survival probabilities through the standard normal CDF, then evaluating the bivariate normal CDF at the resulting quantiles with correlation ρ .

Alternative (Bomfim) joint survival via Venn intersection:

$$Q_{rp}(\tau, T_j) = \prod_{i=1}^j \{1 - [\lambda_r(\tau, T_i) + \lambda_p(\tau, T_i) - \lambda_{rp}(\tau, T_i)]\}. \quad (10)$$

Says: uses conditional default probabilities. Agrees with the unconditional copula version (9) to within a few basis points at the relevant time horizons (paper reports gap of [0.1, 26, 57, 89] bp at [1, 2, 3, 4]Y for $\lambda = [0.05, 0.08]$, $\rho = 0.2$); the copula form is used in the empirical work.

3 Derivations & commentary

Commentary

The valuation is intentionally simple: (1) is just a defaultable bond. The work of the paper is in feeding it with properly bootstrapped survival probabilities (from CDS markets) and in layering counterparty risk via a Gaussian copula. The non-trivial empirical fact is the consistent overpricing of all 11 CLNs by an economically meaningful 0.6%–6.4% (raw) or up to 8.5% (with issuer-counterparty risk priced in), which is interpreted as willingness-to-pay / liquidity premium / too-big-to-fail effect.

Commentary

The reduced-form choice is well-justified for a daily pricing exercise. Structural (Merton-style) models require firm-value volatility and balance-sheet inputs that update at most quarterly, giving them very limited daily variation. Reduced-form with CDS-bootstrapped hazards is the natural daily-frequency choice and also captures the credit-market view of risk rather than the fundamental view.

Commentary

The bootstrap (7) from {1, 3, 5, 7, 10}Y CDS quotes is the standard JPMorgan / O’Kane construction. For Nordea CLNs with 5Y final maturity, the survival probability is dominated by $\lambda_{0,1}$, $\lambda_{1,3}$, $\lambda_{3,5}$, with the largest sensitivity to $\lambda_{3,5}$ because the principal is repaid at maturity (durations confirm this — mostly the 5Y bucket).

Commentary

The Gaussian copula choice ducks the well-known tail-dependence issue (Li 2000; the copula that “broke the world” in 2008). The authors acknowledge that the Clayton copula — with stronger lower-tail dependence — would be more conservative for stress scenarios, but argue that the short sample (one year of post-crisis data) cannot support a more elaborate model. For first-pass validation work the Gaussian copula is fine, but stress-testing should be re-done under a Clayton or t-copula.

Commentary

The empirical finding that theoretical prices “mirror the 5Y CDS spread almost perfectly” is the strongest validation of the model specification. The interest-certificate structure removes interest-rate sensitivity by construction (floating coupon plus quarterly resets), leaving only credit risk. A daily-frequency match between modelled price changes and the 5Y CDS spread says the model is doing what it should; the mispricing level is then a market anomaly, not a model error.

Commentary

Default-correlation estimation (9): rather than equity-return correlation (the common practice, Hull & White 2004), the authors use 5Y-CDS-spread correlations because (i) one reference name (Stena) is not publicly traded and (ii) other reference entities trade in multiple currencies, introducing FX contamination. CDS quotes are all EUR-denominated. Estimated correlations are in [0.13, 0.26], with the largest for ArcelorMittal (0.26) and Volkswagen (0.23).

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

Quantities the paper defines:

- CLN value V^{CLN} via (1) under a risk-free issuer.
- Forward rate f_{t_1, t_2} via (2) for floating-coupon construction.
- Survival $Q(t_V, T)$ via the reduced-form hazard model (3) and its piecewise-flat realisation (7).
- CDS premium and protection legs (4)–(5) and the breakeven CDS spread (6).
- Yield curve via Nelson-Siegel-Svensson (8).
- Joint survival under counterparty risk: Gaussian copula (9) and Bomfim alternative (10).

Canonical empirical setup (§5):

- 11 CLNs issued by Nordea (Apr 2012 – Apr 2013), 5Y maturity, quarterly coupons (3M STIBOR + reference-specific spread).
- Reference entities: ArcelorMittal, Carlsberg, Ericsson, Metso, Nokia, Stena, StoraEnso, Telefonica, UPM-Kymmene, Volkswagen, Volvo.
- Coupon spreads range from STIBOR + 2.05% (Metso) to STIBOR + 5.45% (Stena).
- CDS spreads at {1, 3, 5, 7, 10}Y from Bloomberg / Reuters / FactSet.
- Baseline recovery $R = 40\%$.
- Stressed recovery $R = 10\%$ (post-crisis worst case, Rathgeber-Wang 2011).
- Discount curve: STIBOR + Swedish swap zero curve fitted via NSS (8).
- Default correlations $\rho \in [0.13, 0.26]$ estimated from 5Y CDS time series.

Expected headline numerical findings:

Quantity	Value	Source
Overpricing range, $R = 40\%$, no counterparty risk	0.6%–6.4%	§6.1
Overpricing range, with counterparty (Nordea) credit risk	up to 8.5%	§6.3
Default-correlation range across 11 names	[0.13, 0.26]	Tab. I
CDS-discretisation error, $M = 4$, $r = 2\%$, $S = 500$ bp	$\sim 0.25\%$ relative	p. 15
Copula vs. Bomfim joint-survival gap at 4Y, $\lambda \in \{5\%, 8\%\}$, $\rho = 0.2$	~ 89 bp	p. 22

Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(1)	V^{CLN}	CLN value, risk-free issuer case
(2)	f_{t_1, t_2}	forward rate from spot rates
(3)	λ, Q	Jarrow-Turnbull hazard / survival
(4)–(5)	premium / protection PV	CDS pricing components (O’Kane-Turnbull)
(6)	$S(t_V, t_N)$	breakeven CDS spread
(7)	$Q(t_V, T)$	piecewise-flat hazard from $\{1, 3, 5, 7, 10\}Y$ bootstrap
(8)	$y(\tau)$	NSS yield curve interpolation
(9)	Q_{rp}^C	joint survival under Gaussian copula
(10)	Q_{rp}	joint survival via Venn intersection (Bomfim)

Invariants and identities to verify:

1. *Risk-free limit*: as $R \rightarrow 1$ and $\lambda \rightarrow 0$ ($Q \equiv 1$), (1) reduces to a plain coupon-bond PV: $\sum_i Z(t, T_i)C F + Z(t, T_N)F$.
2. *Bootstrap closure*: (6) evaluated at the bootstrapped hazards should reproduce the input CDS spreads at $\{1, 3, 5, 7, 10\}Y$ exactly (by construction).
3. *Hazard piecewise structure*: (7) is continuous in τ across the boundary points $\{1, 3, 5, 7, 10\}Y$; check that the survival is monotonically decreasing in τ for any non-negative λ vector.
4. *Discretisation error scaling*: as M increases, the protection-leg sum in (5) approaches the integral (5); the relative error is $O(1/M)$.
5. *Copula limits*: at $\rho = 0$ (independence), the Gaussian-copula joint survival in (9) equals the product $Q_r \cdot Q_p$. At $\rho = 1$, it equals $\min(Q_r, Q_p)$.
6. *NSS endpoints*: $y(\tau) \rightarrow \beta_1$ as $\tau \rightarrow \infty$ and $y(\tau) \rightarrow \beta_1 + \beta_2$ as $\tau \rightarrow 0$. Both limits should be recovered by the NSS implementation.
7. *CDS-spread tracking*: empirical correlation between model-implied CLN price changes and observed 5Y CDS spread changes should be near -1 for the reference names where the empirical match is reported as tight (Nokia, Stena, ArcelorMittal).
8. *Two joint-survival agreements*: (9) and (10) agree at $\rho = 0$ exactly; the gap at $\rho = 0.2$ should match the order-of-magnitude figure (89 bp at 4Y for $\lambda \in \{5\%, 8\%\}$).
9. *Counterparty-risk monotonicity*: at fixed reference hazards, increasing ρ increases the joint survival probability (lowest-tranche effect noted by Bomfim), hence *decreases* the counterparty-risk impact on V^{CLN} .

5 Glossary of symbols (paper's notation)

V^{CLN}	Theoretical CLN value at t
C	Coupon rate (constant in eq. 1; floating in practice)
F	CLN notional
R	Recovery rate
T_i, T_N	Coupon dates ($i = 1, \dots, N$), maturity
$Z(t, T_i)$	Risk-free discount factor (STIBOR-based)
$Q(t, T_i)$	Survival probability of reference entity
f_{t_1, t_2}	Forward rate from spot rates r_1, r_2
$\lambda(t)$	Hazard rate
$\lambda_{i,j}$	Piecewise-flat hazard in maturity bucket $[i, j]Y$
$S(t_0, t_N)$	Contractual CDS spread, maturity t_N
$\Delta(t_{n-1}, t_n)$	Day-count fraction
M	Protection-leg discretisation steps per year
$\beta_1, \dots, \beta_4$	NSS yield-curve parameters
λ_1, λ_2	NSS exponential decay parameters
Φ, Φ_Σ	Univariate / bivariate normal CDF
ρ	Default correlation (5Y CDS-spread correlation)
Q_{rp}^C	Gaussian-copula joint survival
Q_{rp}	Bomfim conditional-default joint survival
r, p	Reference entity / counterparty (issuer) subscripts
STIBOR	Stockholm Interbank Offered Rate, Swedish Libor analogue
U_1, U_2	Uniform random variables for copula construction

TODO / Open questions

- Implement the CDS-spread bootstrap (6) $\rightarrow$ piecewise-flat hazards (7) on a sample input (e.g. Nokia 2012 CDS quotes); verify that the bootstrapped hazards reproduce the input spreads exactly.
- Implement the CLN valuation (1) for one Nordea CLN (e.g. Nokia, 3M STIBOR + 3.35%, maturity 2017-04-20) and verify the result tracks the 5Y CDS spread daily.
- Implement NSS (8) and fit to a sample STIBOR + swap zero curve; cross-check against the Riksbank's published rates.
- Compare Gaussian copula (9) to Bomfim's conditional approach (10) on $\lambda \in \{5\%, 8\%\}$, $\rho = 0.2$, time horizon 4Y; target a gap of ~ 89 bp in joint survival probability.
- Sensitivity analysis: reproduce the overpricing range 0.6%–6.4% (baseline) and $\leq 8.5\%$ (with counterparty risk). The discrepancy is the headline empirical finding.
- Recovery-rate sensitivity: re-do the valuation at $R = 10\%$ (post-crisis worst case from Rathgeber-Wang 2011) to bound the model risk on the recovery assumption.
- Robustness: re-do with CDS spreads from a different data source (the paper checks Bloomberg vs. Reuters vs. FactSet); should not move the headline conclusion materially.